Mengke Zhao

mkzhao@nju.edu.cn | meng-ke.github.io | meng-ke | Nanjing University, Nanjing, China

Research Profile

I study how nonlinear physical interactions generate structure across scales, and how that structure can reveal the hidden states and dynamics of complex systems. The interstellar medium is my primary natural laboratory, where gravity, turbulence, magnetic fields, radiation, and feedback form a strongly coupled multiphysics system; the same questions and methods extend naturally to turbulent MHD and fluid systems.

My research connects three directions:

- **Multiscale physics and nonlinear coupling:** MHD scaling relations, magnetic regulation, scale-dependent morphology, and the gradual $B - \rho$ transition across physical regimes.
- **3D structure and gravitational dynamics:** Volume Density Mapper (VDM), Multiscale Decomposition Reconstruction (MDR), local multi-fractal density scaling, and gravitational tidal fields.
- **AI4Science and representation learning:** scale-aware diagnostics for generative models and ScaleAware-JEPA for learning latent representations of physical dynamical states.

Education & Employment

School of Astronomy and Space Science, Nanjing University — *Nanjing, China*
Postdoctoral Researcher at Nanjing University — 2024-present

South-Western Institute for Astronomy Research, Yunnan University — *Kunming, China*
Visiting Scholar, Yunnan University — 2023-2024

Xinjiang Astronomical Observatory, Chinese Academy of Sciences — *Urumqi, China*
Ph.D. in Astrophysics (2020–2024); M.S. in Astrophysics (2018–2020) — 2018–2024

Articles & Manuscripts

2026 — Current Research Directions

Zhao, Mengke, Li, Guang-Xing, & Qiu, Keping. *Morphological Identification of Dynamical Regimes in MHD Turbulence with ScaleAware-JEPA*. arXiv preprint (2026). [arXiv]

Zhao, Mengke, Li, Guang-Xing, Qiu, Keping, & Li, Shanghuo. *Scale-Vector Alignment: A Scale-Aware Framework for Spatially Resolved Morphological Similarity in Astronomical Images*. Submitted to *ApJS* (2026). [arXiv] [Code]

Zhao, Mengke, Li, Guang-Xing, Qiu, Keping, & Zeng, Guangya. *Gravity-driven Emergence of Multi-fractal Density Structure in the Orion A Integral Shaped Filament*. Submitted to *ApJ* (2026). [arXiv]

Zhao, Mengke, Li, Guang-Xing, & Qiu, Keping. *A Unified Magnetohydrodynamic Scaling Relation for the Multiphase Interstellar Medium*. Published in *The Astrophysical Journal* (2026). [arXiv]

Zhao, Mengke, Li, Guang-Xing, Xu, Duo, & Qiu, Keping. *Scale-Aware Adversarial Analysis: A Diagnostic for Generative AI in Multiscale Complex Systems*. Under review at *Machine Learning: Science and Technology* (2026). [arXiv]

Selected Publications & Foundational Work

Zhao, Mengke, Qiu, Keping, Kang, Ji-hyun, Tang, Xindi, et al. *BISTRO Survey: Gravity-Dominated and Magnetically Regulated Star Formation in M17 SW*. *The Astrophysical Journal*, 1005, 91 (2026). [DOI]

Zhao, Mengke, Li, Guang-Xing, Xu, Duo, & Qiu, Keping. *Equation versus AI: Predicting Density and Measuring Width of Molecular Clouds by Multiscale Decomposition*. *The Astrophysical Journal*, 997, 345 (2026). [DOI] [Code]

Li, Guang-Xing, & **Zhao, Mengke**. *Volume Density Mapper: 3D Density Reconstruction Algorithm for Molecular Clouds*. arXiv preprint (2025). [arXiv] [Code]

Li, Guang-Xing, & **Zhao, Mengke**. *Magnetic, Kinetic, and Transition Regime: Spatially Segregated Structure of Compressive MHD Turbulence*. *Monthly Notices of the Royal Astronomical Society*, 542, 3246–3252 (2025). [DOI]

Zhao, Mengke, Li, Guang-Xing, & Qiu, Keping. *Slope of Magnetic Field–Density Relation as an Indicator of Magnetic Dominance*. *The Astrophysical Journal*, 976, 209 (2024). [ADS]

Earlier Work on Magnetic Fields & the Velocity Gradient Technique

Zhao, Mengke, Jianjun Zhou, Yue Hu, A. Lazarian, Xindi Tang, et al. *Magnetic Field of Molecular Gas Measured with the Velocity Gradient Technique I: Orion A*. *The Astrophysical Journal*, 934, 45 (2022). [DOI]

Zhao, Mengke, Guang-Xing Li, Jianjun Zhou, Xindi Tang, Jarken Esimbek, et al. *Magnetic Fields in Giant Filaments Probed by the Velocity Gradient Technique: Regular Magnetic Field Interrupted by Magnetization Gaps*. *The Astrophysical Journal*, 961, 124 (2024). [arXiv]

Zhao, Mengke, Jianjun Zhou, Willem A. Baan, Yue Hu, A. Lazarian, et al. *Magnetic Field of Molecular Gas Measured with the Velocity Gradient Technique II: Curved Magnetic Field in a kpc-Scale Bubble of NGC 628*. *The Astrophysical Journal*, 966, 124 (2024). [DOI]

Research Software

Scale-Vector Alignment (SVA) — scale-aware, spatially resolved morphological comparison for multiscale images. [GitHub]

Volume Density Mapper / MDR — 3D density reconstruction, characteristic-scale estimation, and projected volume-density mapping from column-density data. [GitHub]

Curvature Mapping Method — Python package for magnetic-field curvature and projected Lorentz-force mapping from polarization data. [GitHub] [PyPI]

Grants & Observational Programs

PI, National Natural Science Foundation of China (NSFC) Young Scientists Fund (Grant No. 12503029), 2026–2028.

Core Member, B-fields In STar-forming Region Observations (BISTRO) Survey; active contributor to the Second General Program (PI: K. Pattle) and the 3rd HMPO Tiger Team (PI: K. Qiu).

PI, JCMT Cycle 21B Observational Program, SCUBA-2/POL-2 (Band 1).

Co-PI, JCMT Cycle 22A Observational Program (with Prof. Jian-Jun Zhou).